

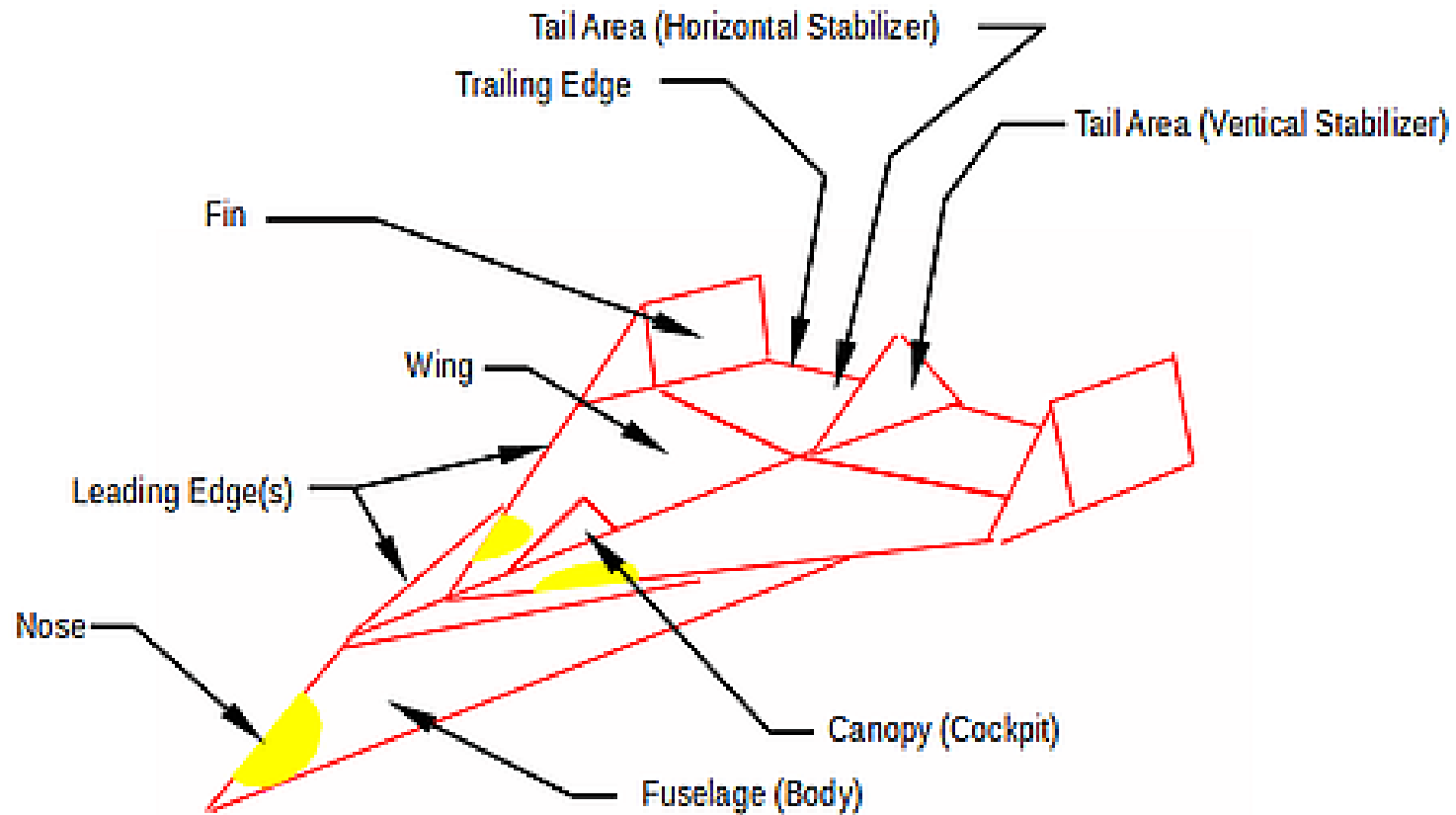
Design of Experiments and Gear Mechanisms

Lecture 11

Outline

- What is robust design?
- Design of Experiments
- Design of Gears and Gearboxes
- Classroom activity:
 - Creating a simple gearbox using Fusion 360

Paper Airplane



Goals for Designed Experiments

- Modeling
 - Understanding relationships between design parameters and product performance
 - Understanding effects of noise factors
- Optimizing
 - Reducing product or process variations
 - Optimizing nominal performance

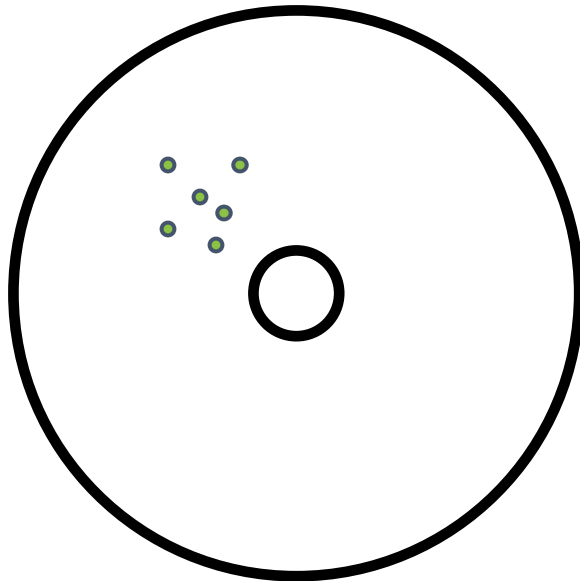
Robust Designs

A **robust product or process** performs correctly, even in the presence of noise factors.

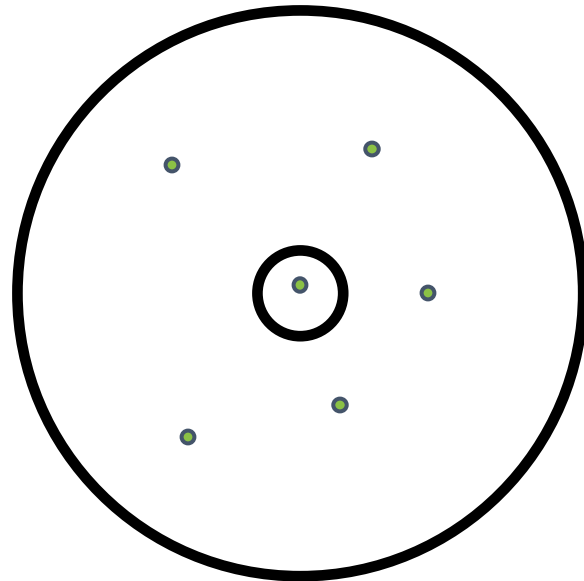
Noise factors may include:

- parameter variations
- environmental changes
- operating conditions
- manufacturing variations

Who is the better target shooter?

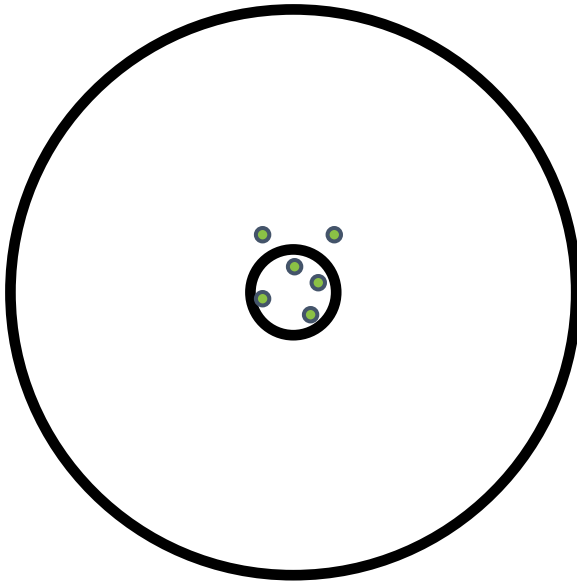


Sam



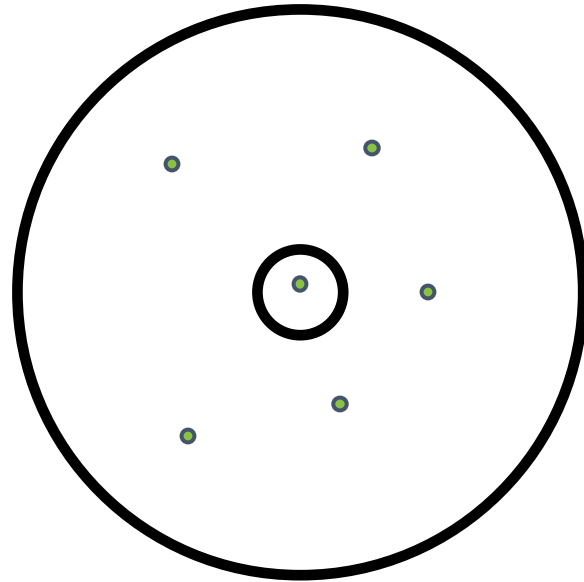
John

Who is the better target shooter?



Sam

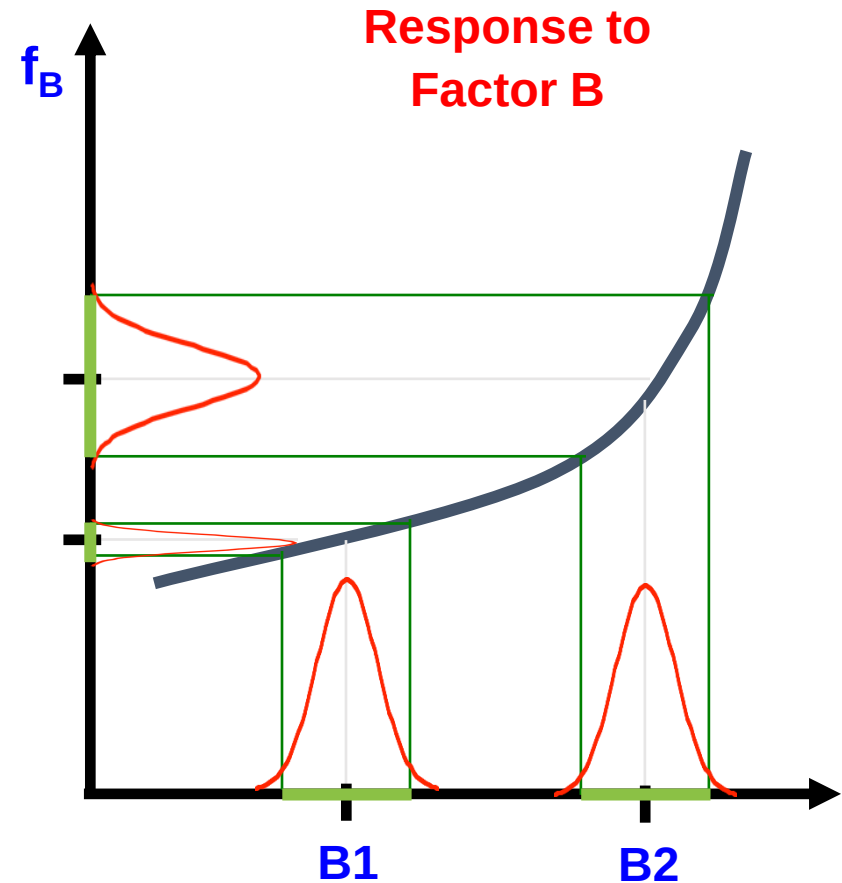
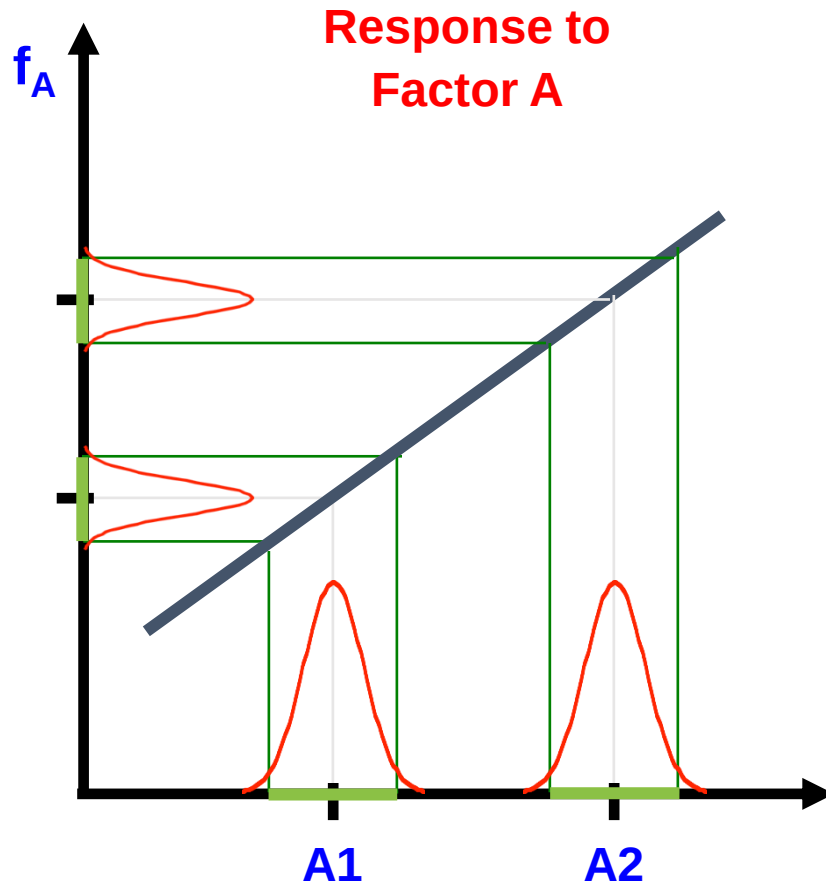
Sam can simply
adjust his sights.



John

John requires
lengthy training.

Exploiting Non-Linearity to Achieve Robust Performance



$$\text{Response} = f_A(A) + f_B(B)$$

What level of factor B gives the robust response?

How do we use factor A?

Robust Design Procedure

Step 1: Parameter Diagram

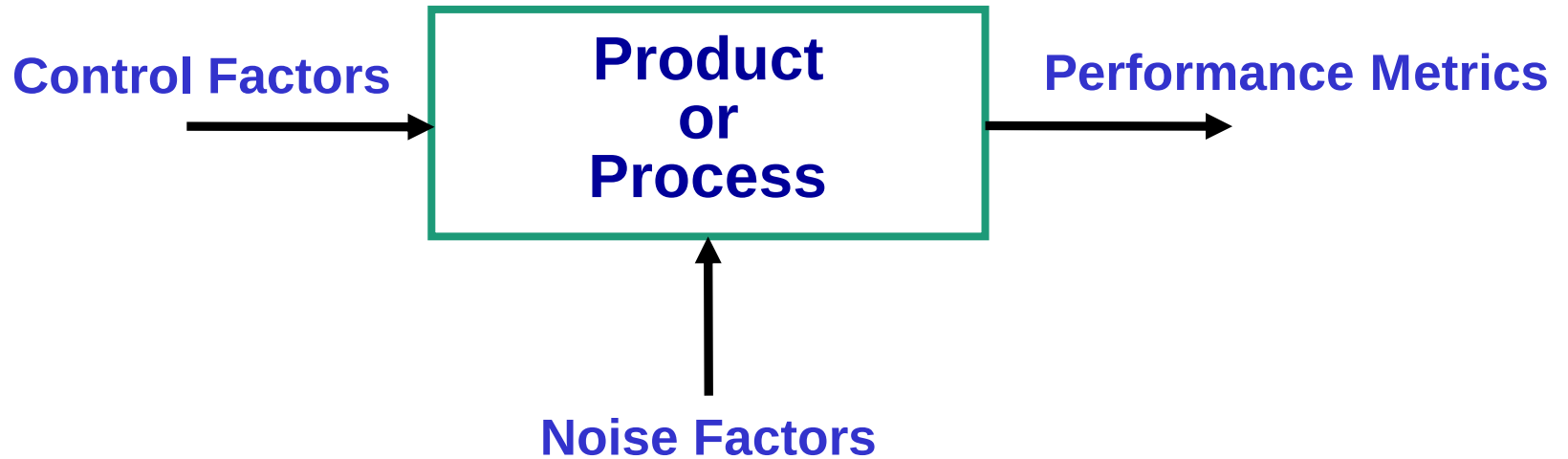
Step 1: Select appropriate controls, response, and noise factors to explore experimentally.

- Control factors (input parameters)
- Noise factors (uncontrollable)
- Performance metrics (response)

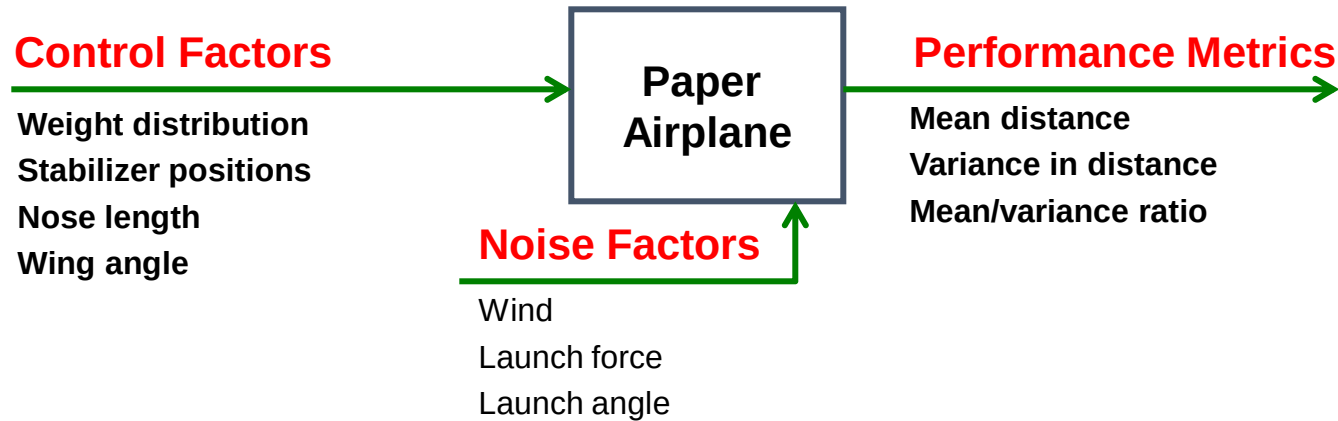
Example: Brownie Mix

- Control Factors
 - Recipe Ingredients (quantity of eggs, flour, chocolate)
 - Recipe Directions (mixing, baking, cooling)
 - Equipment (bowls, pans, oven)
- Noise Factors
 - Quality of Ingredients (size of eggs, type of oil)
 - Following Directions (stirring time, measuring)
 - Equipment Variations (pan shape, oven temp)
- Performance Metrics
 - Taste Testing by Customers
 - Sweetness, Moisture, Density

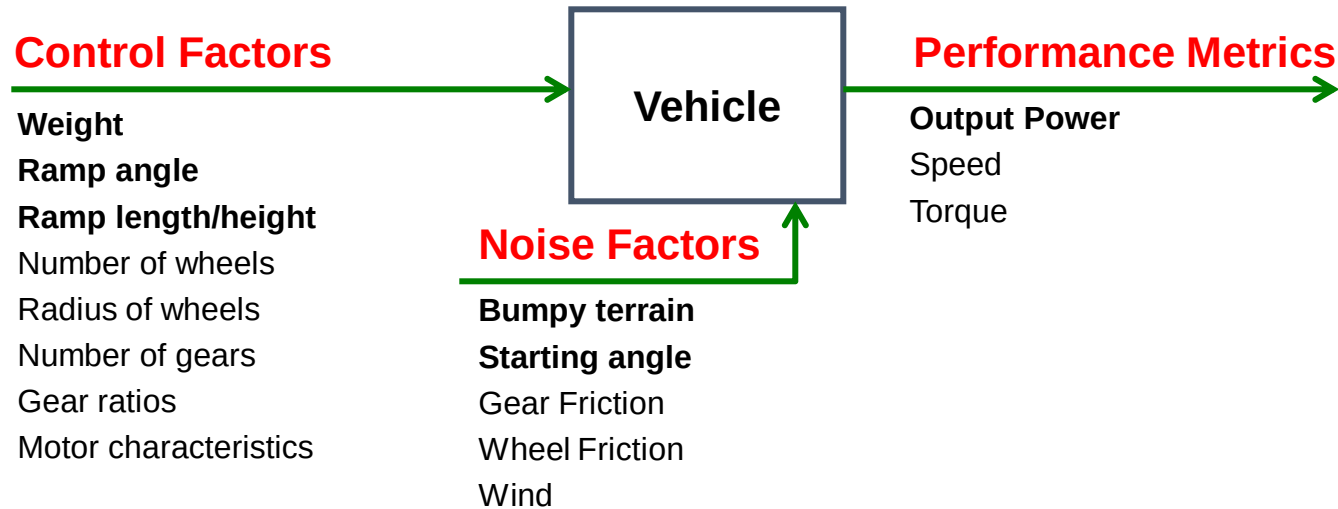
The “P” Diagram



Parameter Diagram for Paper Airplane



Parameter Diagram for Vehicle



Robust Design Procedure

Step 2: Objective Function

Step 2: Define an objective function (of the response) to optimize.

- maximize desired performance
- minimize variations
- target value
- signal-to-noise ratio

Types of Objective Functions

Larger-the-Better

e.g. performance

$$\eta = \mu^2$$

Smaller-the-Better

e.g. variance

$$\eta = 1/\sigma^2$$

Nominal-the-Best

e.g. target

$$\eta = 1/(\mu - t)^2$$

Signal-to-Noise

e.g. trade-off

$$\eta = 10\log[\mu^2/\sigma^2]$$

Robust Design Procedure

Step 3: Plan the Experiment

Step 3: Plan experimental runs to elicit desired effects.

- Use full or fractional factorial designs to identify interactions.
- Use an orthogonal array to identify main effects with minimum of trials.
- Induce variance to see the effects of noise factors using natural variation, compounded noise, or outer arrays.

Experiment Design: Full Factorial

- Consider k factors, n levels each.
- Test all combinations of the factors.
- The number of experiments is n^k .
- Generally this is too many experiments, but we are able to reveal all of the interactions.

Expt #	Param A	Param B
1	A1	B1
2	A1	B2
3	A1	B3
4	A2	B1
5	A2	B2
6	A2	B3
7	A3	B1
8	A3	B2
9	A3	B3

2 factors, 3 levels each:

$$n^k = 3^2 = 9 \text{ trials}$$

4 factors, 3 levels each:

$$n^k = 3^4 = 81 \text{ trials}$$

Experiment Design: One Factor at a Time

- Consider k factors, n levels each.
- Test all levels of each factor while freezing the others at nominal level.
- The number of experiments is $(n-1)k+1$.
- BUT this is an unbalanced experiment design.

Expt #	Param A	Param B	Param C	Param D
1	A2	B2	C2	D2
2	A1	B2	C2	D2
3	A3	B2	C2	D2
4	A2	B1	C2	D2
5	A2	B3	C2	D2
6	A2	B2	C1	D2
7	A2	B2	C3	D2
8	A2	B2	C2	D1
9	A2	B2	C2	D3

4 factors, 3 levels each:

$$(n-1)k+1 =$$

$$(3-1) \times 4 + 1 = 9 \text{ trials}$$

Experiment Design: Orthogonal Array

- Consider k factors, n levels each.
- Test all levels of each factor in a balanced way.
- The number of experiments is order of $1+k(n-1)$.
- This is the smallest balanced experiment design.
- BUT main effects and interactions are confounded.

Expt #	Param A	Param B	Param C	Param D
1	A1	B1	C1	D1
2	A1	B2	C2	D2
3	A1	B3	C3	D3
4	A2	B1	C2	D3
5	A2	B2	C3	D1
6	A2	B3	C1	D2
7	A3	B1	C3	D2
8	A3	B2	C1	D3
9	A3	B3	C2	D1

4 factors, 3 levels each:

$$1+k(n-1)=$$

$$1+4(3-1) = 9 \text{ trials}$$

Using Inner and Outer Arrays

- Induce the same noise factor levels for each combination of controls in a balanced manner

4 factors, 3 levels each:
L9 inner array for controls

A1	B1	C1	D1
A1	B2	C2	D2
A1	B3	C3	D3
A2	B1	C2	D3
A2	B2	C3	D1
A2	B3	C1	D2
A3	B1	C3	D2
A3	B2	C1	D3
A3	B3	C2	D1

3 factors, 2 levels each:
L4 outer array for noise

E1	E1	E2	E2
F1	F2	F1	F2
G2	G1	G2	G1

inner x outer =
L9 x L4 =
36 trials

[illegible]

		A1								A2								
		B1				B2				B1				B2				
		C1		C2		C1		C2		C1		C2		C1		C2		
		D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	
E1	F1	G1	x			x		x	x			x	x		x			x
		G2		x	x		x			x	x			x		x	x	
	F2	G1		x	x		x			x	x			x		x	x	
		G2	x			x		x	x			x	x		x			
E2	F1	G1		x	x		x			x	x			x		x	x	
		G2	x			x		x	x			x	x		x			x
	F2	G1	x			x		x	x			x	x		x			
		G2		x	x		x			x	x			x		x	x	x

[illegible]

		A1								A2								
		B1				B2				B1				B2				
		C1		C2		C1		C2		C1		C2		C1		C2		
		D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	D1	D2	
E1	F1	G1	x												x			
		G2				x												x
	F2	G1						x					x					
		G2					x				x							
E2	F1	G1					x				x							
		G2						x					x					
	F2	G1				x												x
		G2	x												x			

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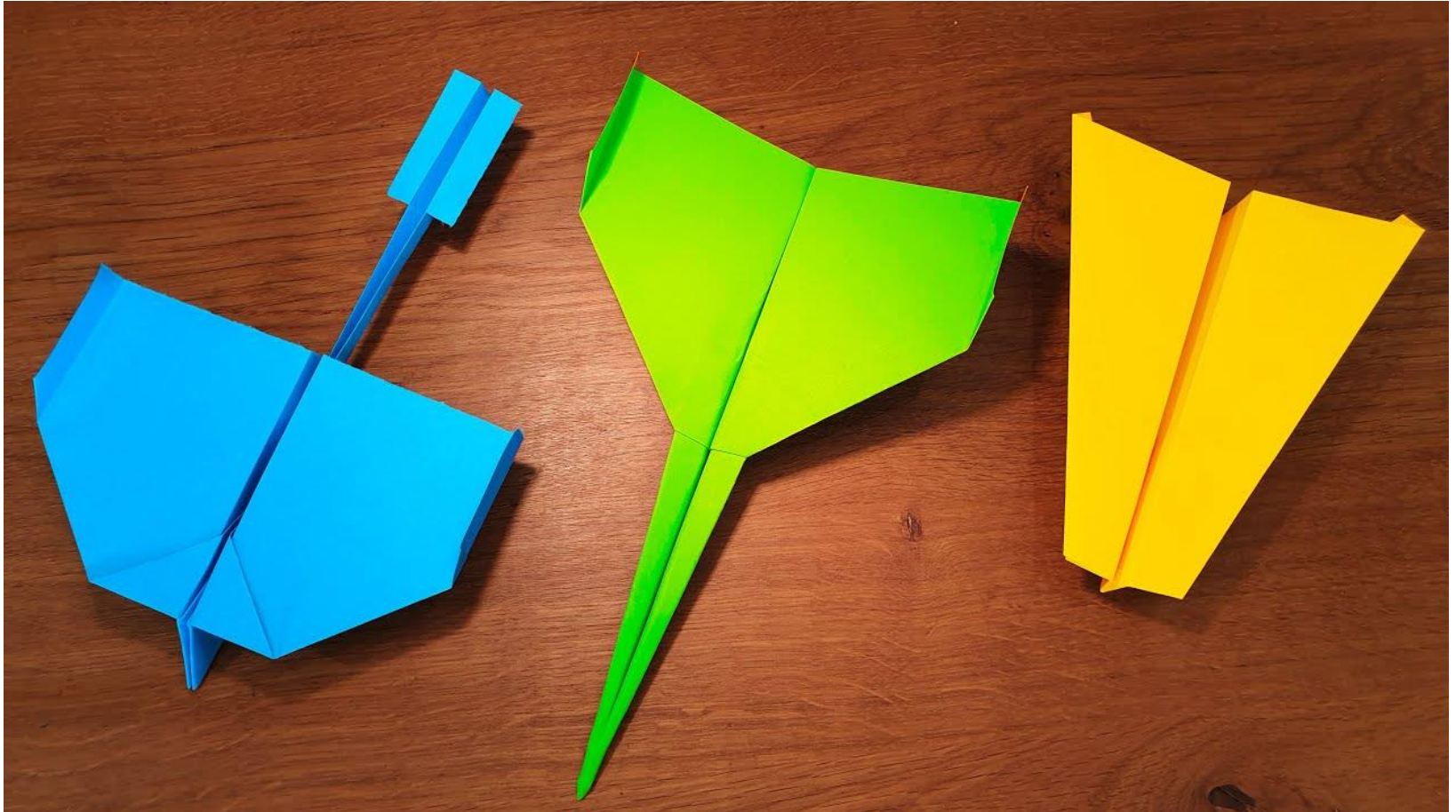
Robust Design Procedure

Step 4: Run the Experiment

Step 4: Conduct the experiment.

- Vary the control and noise factors
- Record the performance metrics
- Compute the objective function

Paper Airplanes



Paper Airplane Experiment

A: weight dist.	B: stabilizer	C: nose length	D: wing angle	Mean Distance	Std Deviation	Variance	Signal/NoiseR
A1	B1	C1	D1	22.3	7.5	56.5	9.4
A1	B2	C2	D2	16.0	4.0	16.0	12.0
A1	B3	C3	D3	17.0	3.2	10.5	14.4
A2	B1	C2	D3	18.6	3.3	10.8	15.1
A2	B2	C3	D1	17.0	5.7	32.0	9.6
A2	B3	C1	D2	22.2	5.1	25.7	12.8
A3	B1	C3	D2	19.4	4.2	17.3	13.4
A3	B2	C1	D3	9.6	4.2	17.8	7.1
A3	B3	C2	D1	23.5	2.5	6.5	19.3

Enter Experimental Data Below					
Experiment #	Throw 1	Throw 2	Throw 3	Throw 4	Throw 5
1	19.5	21	35	15	21
2	21	19	11	14	15
3	13	14	19	19	20
4	18	19	16	16	24
5	23	19	11	11	21
6	28	21	21	15	26
7	25	20	17	21	14
8	13	5	15	7	8
9	20	27	23	23	24.5

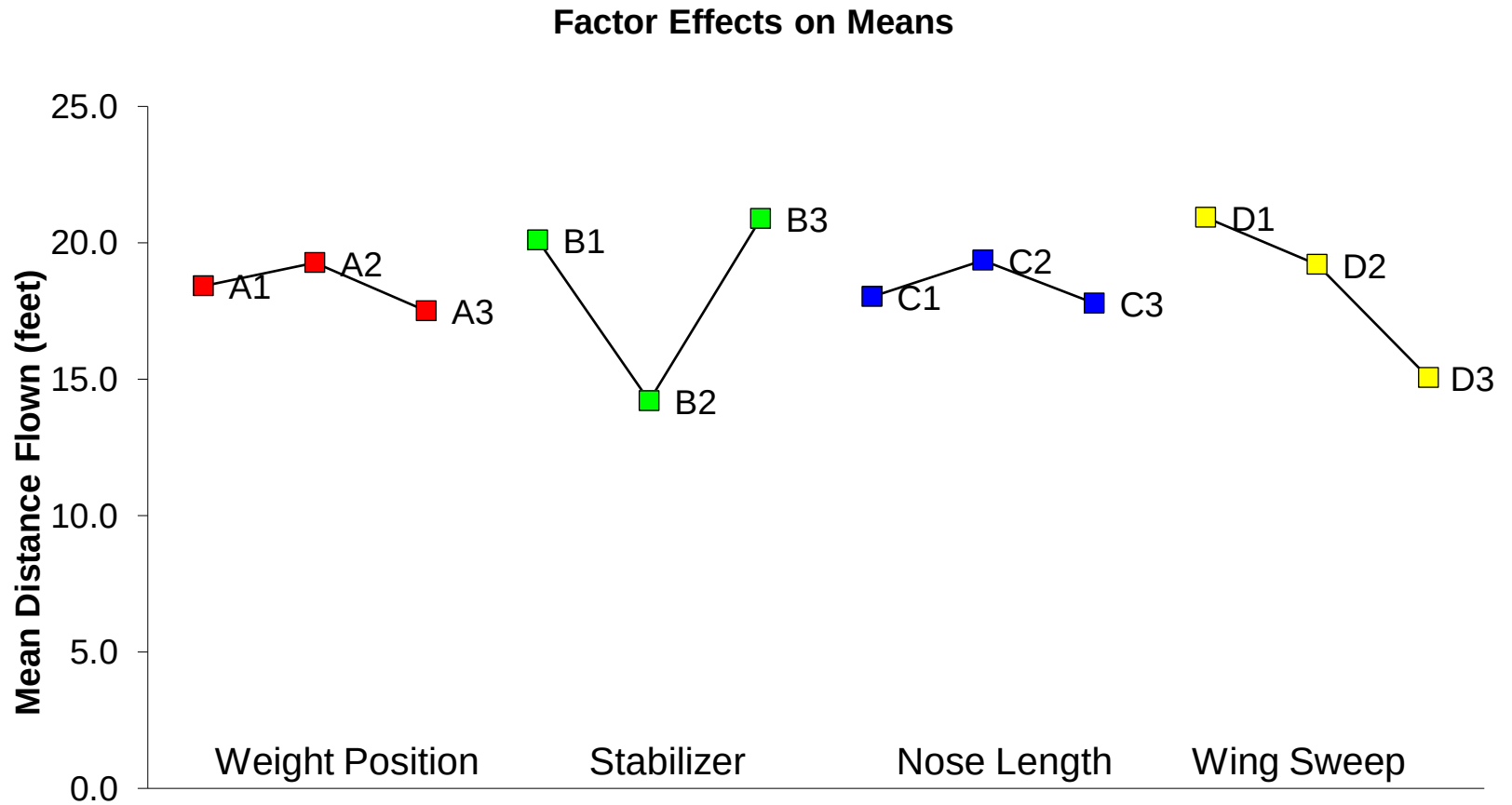
Robust Design Procedure

Step 5: Conduct Analysis

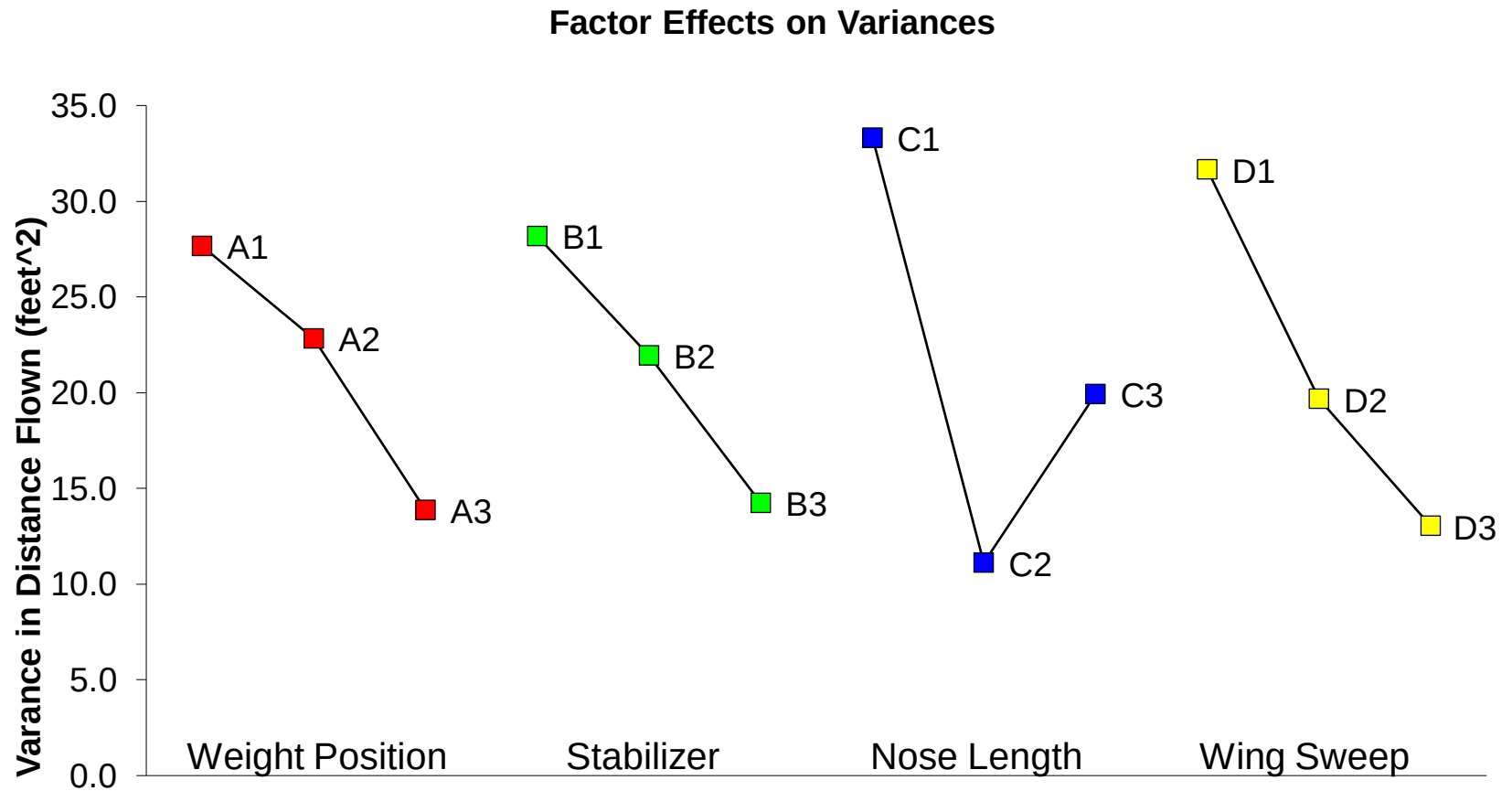
Step 5: Perform analysis of means.

- Compute the mean value of the objective function for each factor setting.
- Identify which control factors reduce the effects of noise and which ones can be used to scale the response. (2-Step Optimization)

Factor Effects Charts



Factor Effects Charts



Robust Design Procedure

Step 6: Select Setpoints

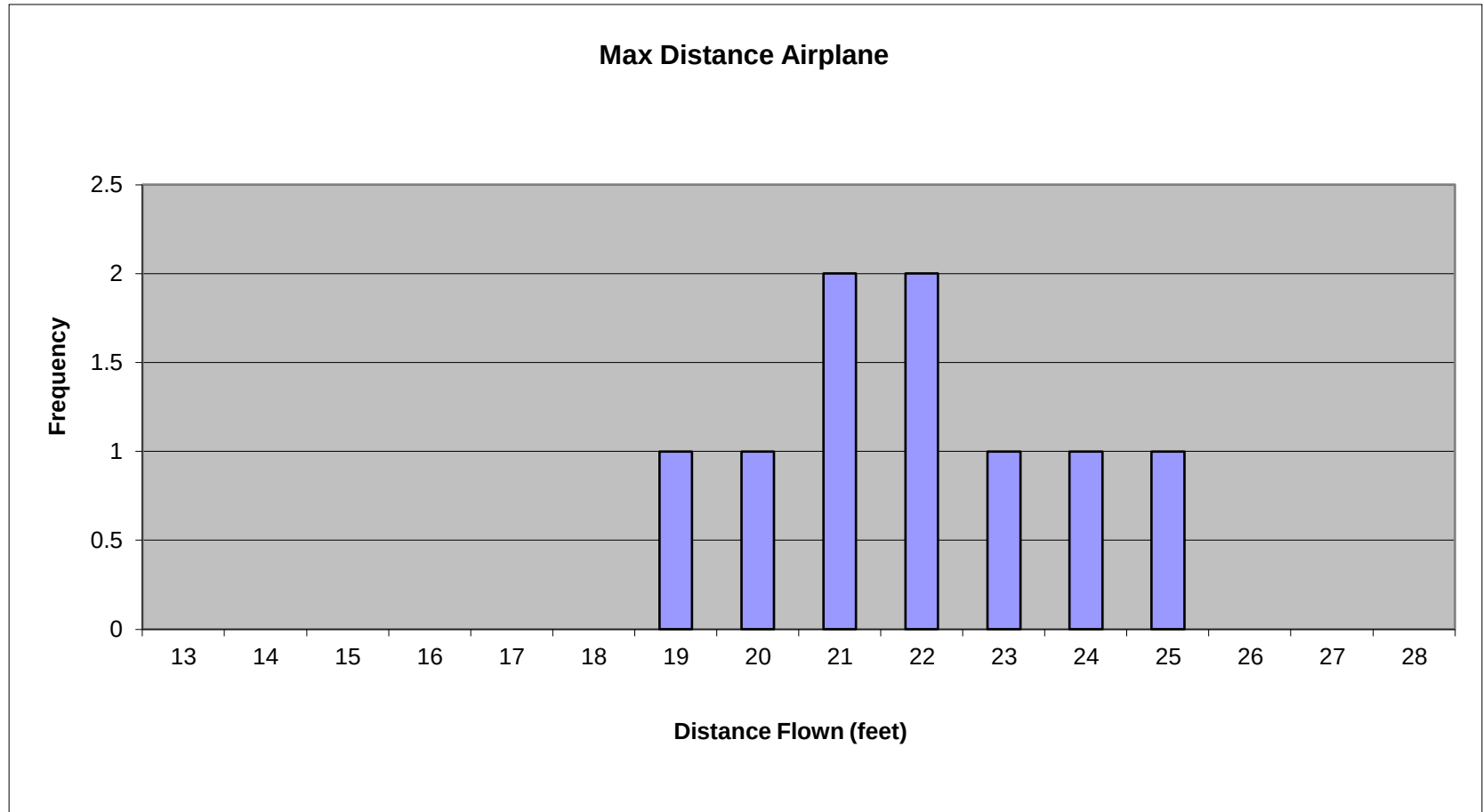
Step 6: Select control factor setpoints.

- Choose settings to maximize or minimize objective function.
- Consider variations carefully. (Use ANOM on variance to understand variation explicitly.)

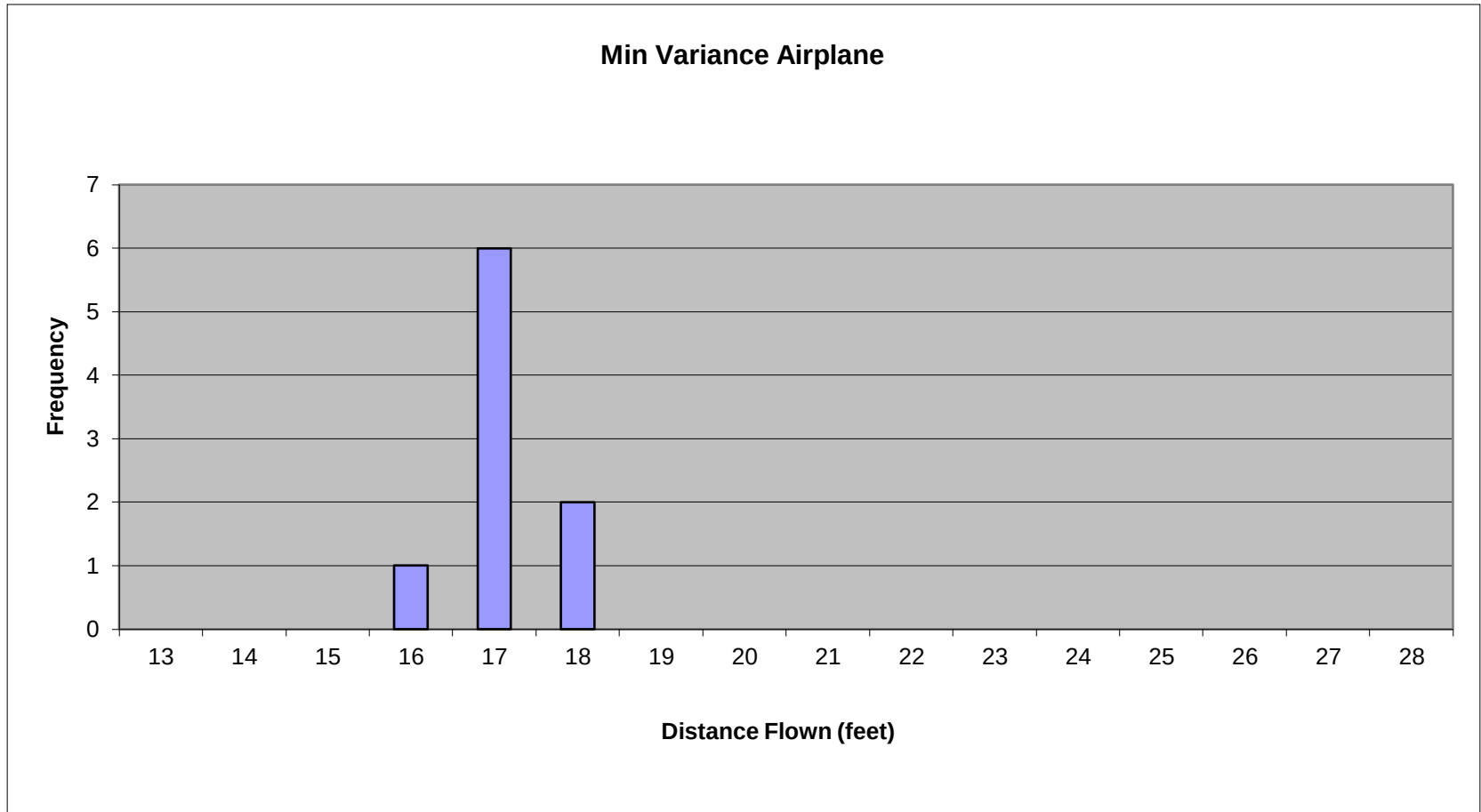
Advanced use:

- Conduct confirming experiments.
- Set scaling factors to tune response.
- Iterate to find optimal point.
- Use higher fractions to find interaction effects.
- Test additional control and noise factors.

Confirmation of setpoints



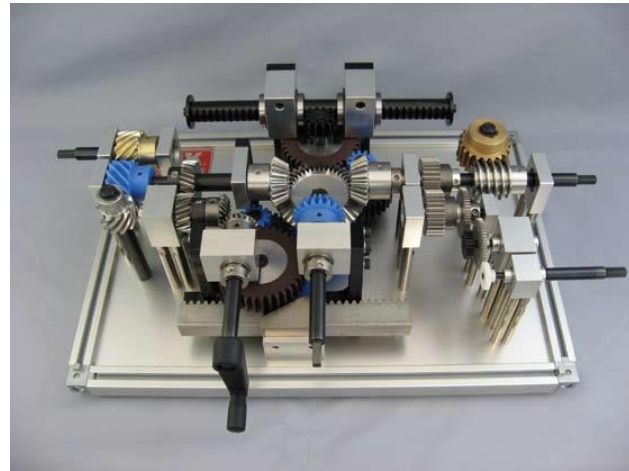
Confirmation of setpoints



Gears and Gear Design

Gears and Gear Design

- Gears is a simple machine that can transmit power/motion from one shaft to another.
 - Speed ratio
 - Torque ratio
 - Change direction



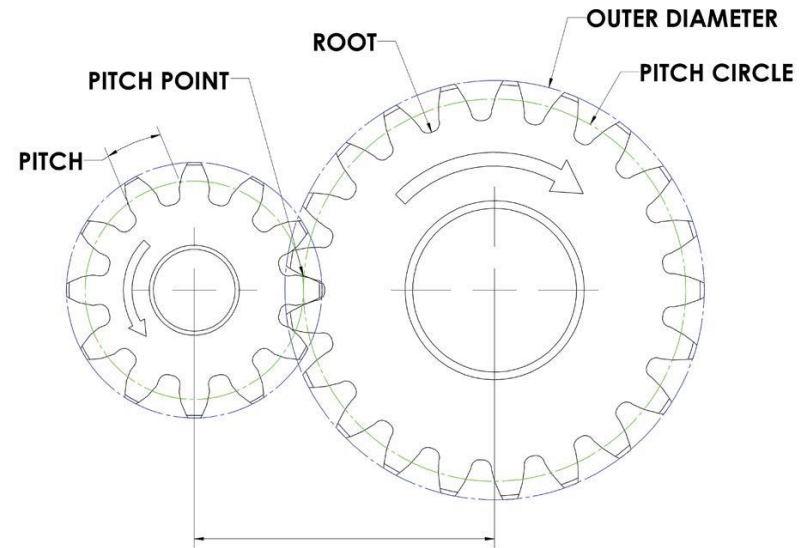
Speed

Torque



Gear Terminology

- **Pitch:** $p = \frac{\pi d}{n}$
 - where d diameter of pitch circle,
 n number of teeth
- **Module:** $m = \frac{d}{n}$
 - Gears mesh only if their
modules are equal



properly meshed gears



gears with modules 0.8 and 2

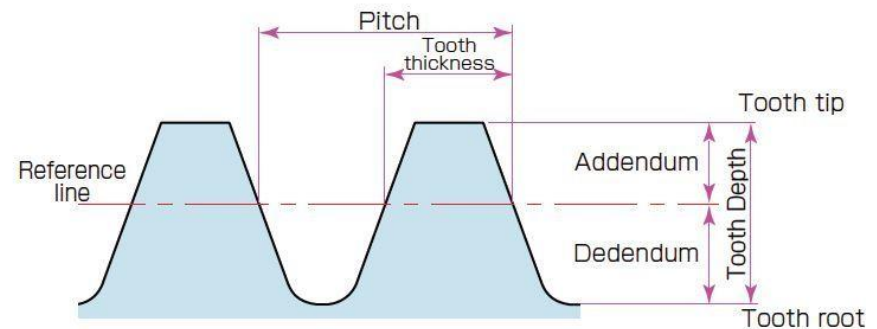
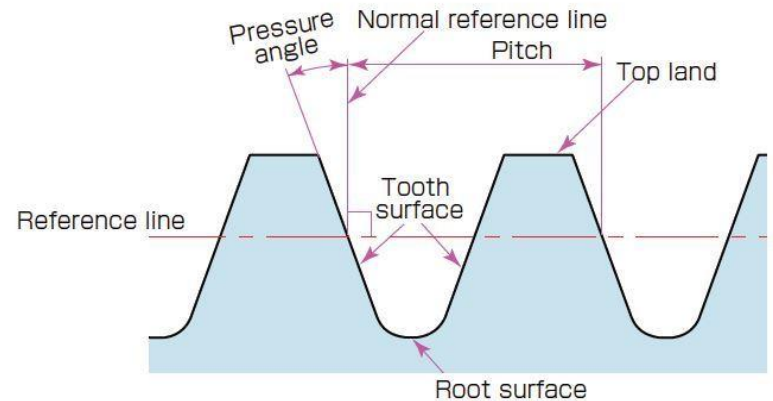
Gear Terminology

- **Pressure Angle**

- Pressure angle is the leaning angle of a gear tooth, an element determining the tooth profile.
- The pressure angle (α) is usually set to 20° , however, 14.5° gears were prevalent

- **Tooth Depth and Thickness**

- Tooth depth is determined from the size of the module m and it is standardized (ISO) –
 - $h = 2.25 * m$
 - $h_a = 1 * m$
 - $h_d = 1.25 * m$
- Tooth thickness is half of pitch p
 - $s = \frac{p}{2} = \frac{\pi m}{2}$



Gear Terminology

- **Reference Diameter**

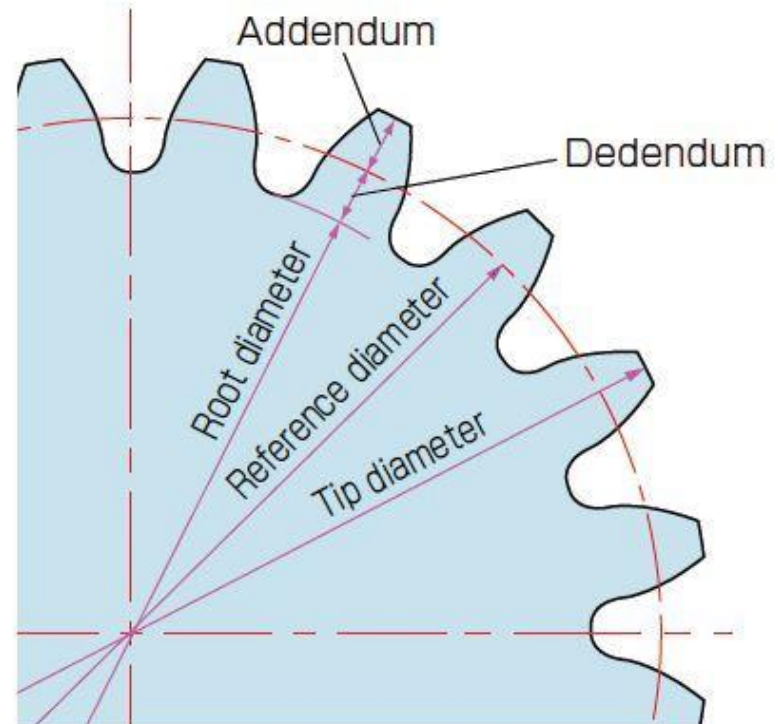
- $d = m * n$

- **Tip Diameter**

- $d_a = d + 2 * m$

- **Root Diameter**

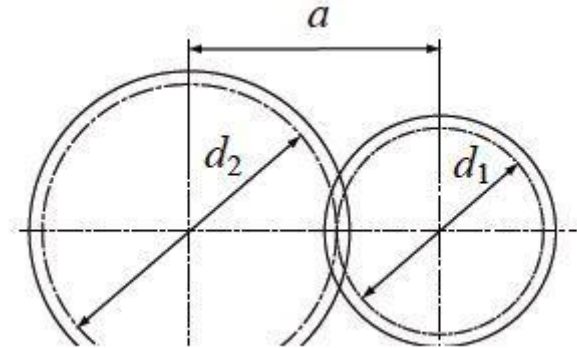
- $d_f = d - 2.5 * m$



Gear Terminology

- **Center Distance**

- $a = \frac{d_1 + d_2}{2}$

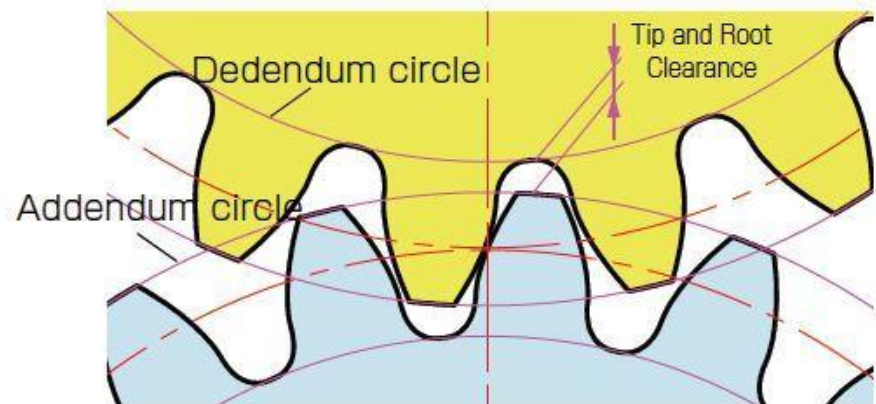


- **Tip to Root Clearance**

- a clearance (play) between tooth surfaces of paired gears in mesh
 - necessary for smooth mating
 - $c = 0.25 m$

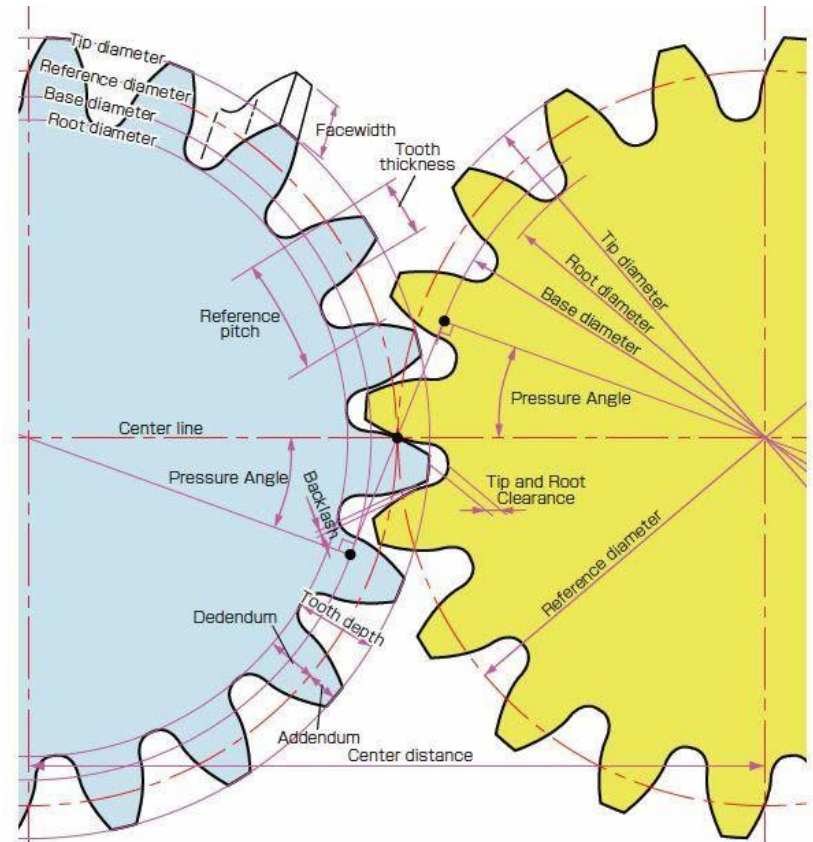
- **Backlash**

- $j = 2 * c * \sin(\alpha)$



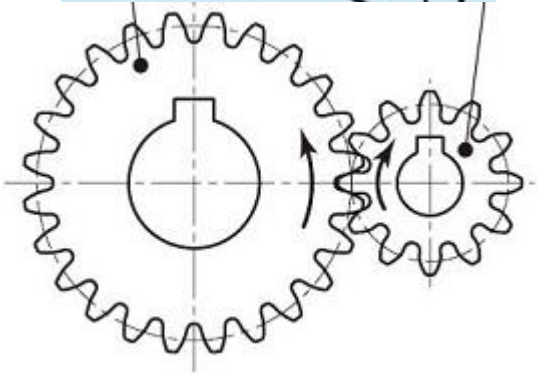
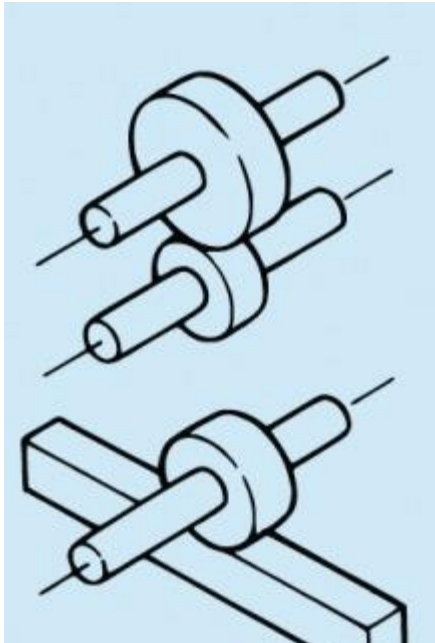
Gear Terminology

Terms	Symbols	Terms	Symbols
Module	m	Tooth Thickness	s
Pressure Angle	α	Reference Diameter	d
Number of Tooth	n	Tip Diameter	d_a
Pitch	p	Root Diameter	d_f
Tooth Depth	h	Center Distance	a
Addendum	h_a	Backlash	j
Dedendum	h_f	Tip and Root Clearance	c

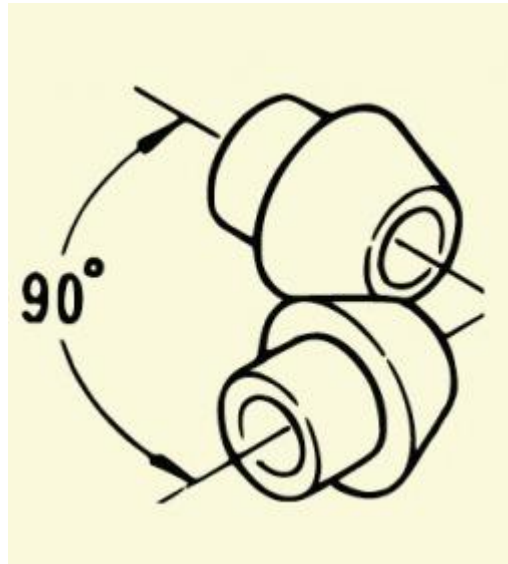


Gear Shafts

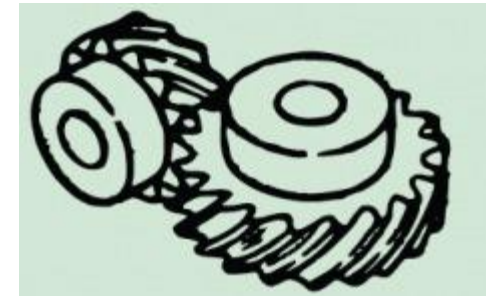
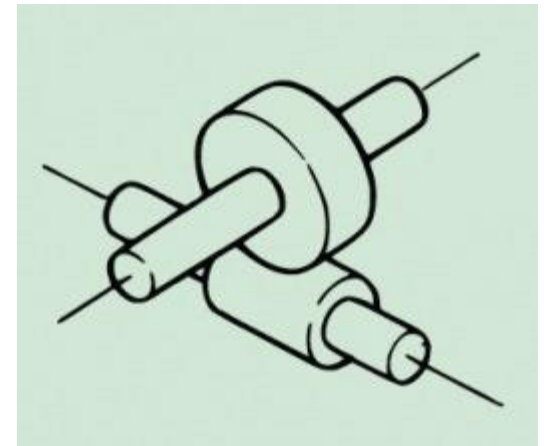
Parallel Axis Gears



Intersecting Gears



Non-Intersecting and Non-parallel Gears



Types of Gears



Spur Gears



Helical Gears



Gear Rack



Bevel Gears



Miter Gears



Worm Gears

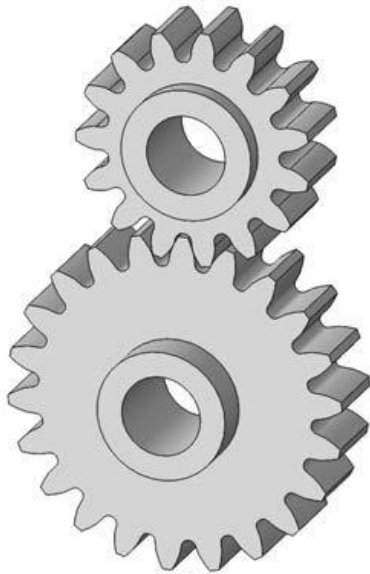


Screw Gears



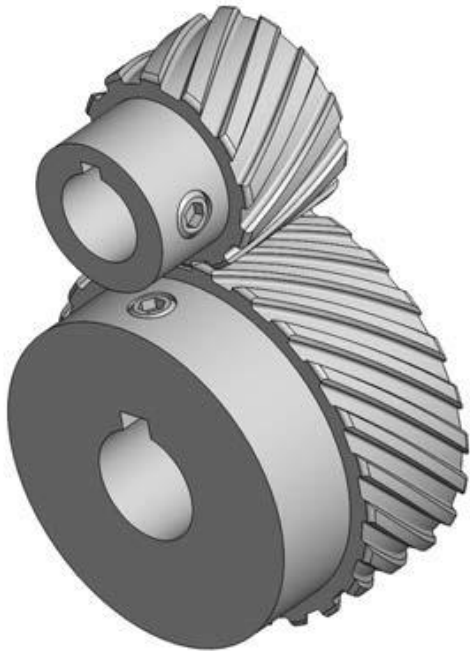
Internal Gears

Spur Gear



- A spur gear is a cylindrical shaped gear, in which the teeth are parallel to the axis.
- The most easiest to manufacture
- No limit in the combination of the number of gear teeth of paired gears
- Driver gear is called *pinion*

Helical Gears



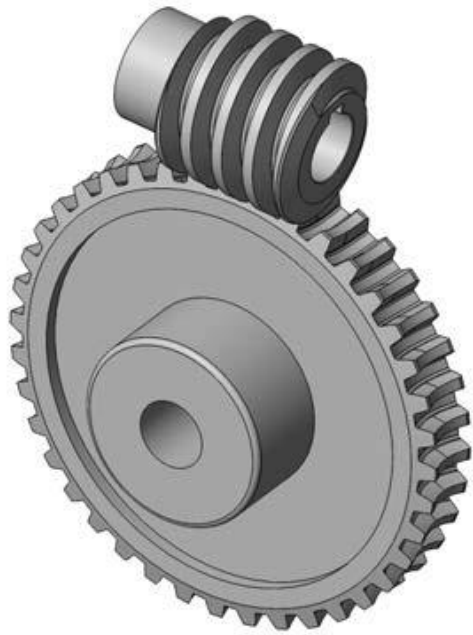
- A helical shape that allows for smoother and quieter operation
- More strength than the spur gear of the same size
- Suitable for use in high speed rotations
- Produces axial thrust force, need to cope with these extra forces
- There is no limit in the combination of number of gear teeth of paired gears.

Bevel Gears



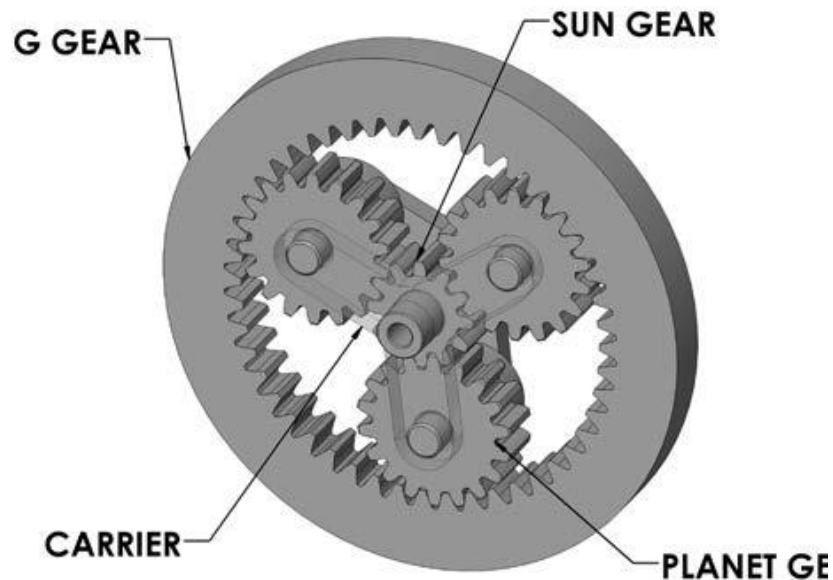
- These gears have teeth that are angled to the gear axis, allowing for the transmission of power between shafts that are at an angle to each other.
- Bevel gears can have different ratios of teeth and can be used to adjust the speed and torque of the output shaft.

Worm Gears



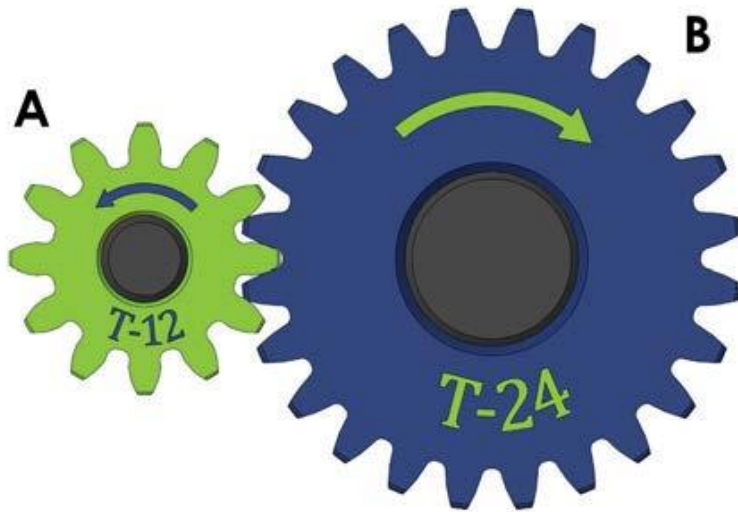
- These consist of a worm gear and a worm wheel, where the worm gear has a screw-like thread that meshes with the teeth on the worm wheel
- Worm gears are used to transmit power between non-parallel and non-intersecting shafts and provide high reduction ratios

Planetary Gears



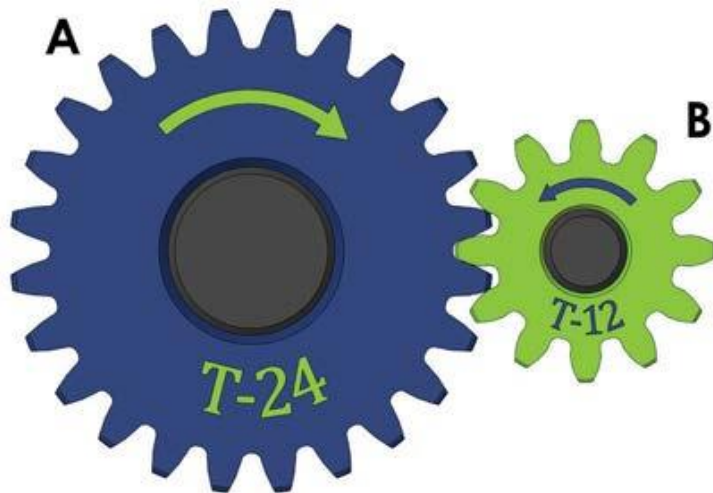
- These gears consist of a central sun gear, several planet gears, and an outer ring gear.
- Planetary gears provide high reduction ratios and are commonly used in automotive and industrial applications.

Gear Ratios



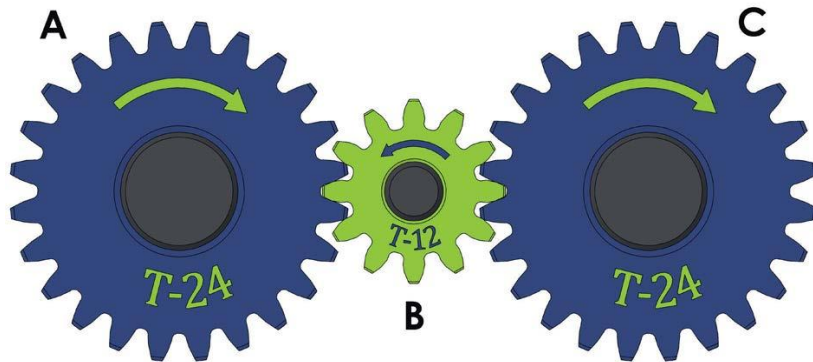
- A is a driver gear and has 12 teeth (aka pinion), B is a driven gear and has 24 teeth.
- Gear Ratio = $\frac{T_{\text{driven}}}{T_{\text{driver}}} = \frac{T_B}{T_A} = \frac{24}{12} = 2:1$
- It means there is a twice reduction in speed and a twice increase in torque. For example, A torque is 1 Nm and 200 rpm => B torque is 2 Nm and 100 rpm.

Gear Ratios



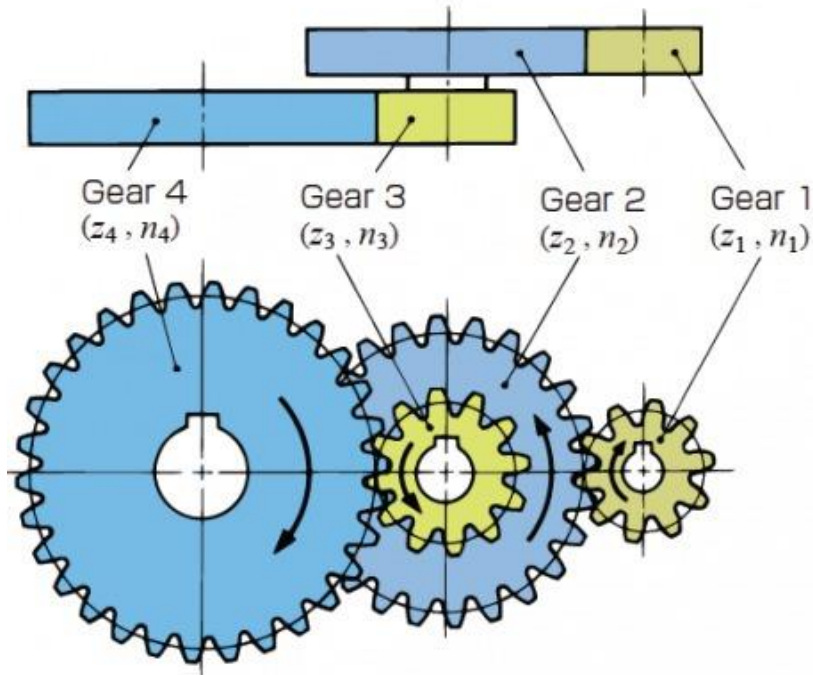
- A is a driver gear and has 24 teeth, B is a driven gear and has 12 teeth.
- Gear Ratio = $\frac{T_{\text{driven}}}{T_{\text{driver}}} = \frac{T_B}{T_A} = \frac{12}{24} = 1:2$
- It means there is a twice reduction in torque and a twice increase in speed. For example, A torque is 1 Nm and 200 rpm => B torque is 0.5 Nm and 400 rpm.

Gear Ratios



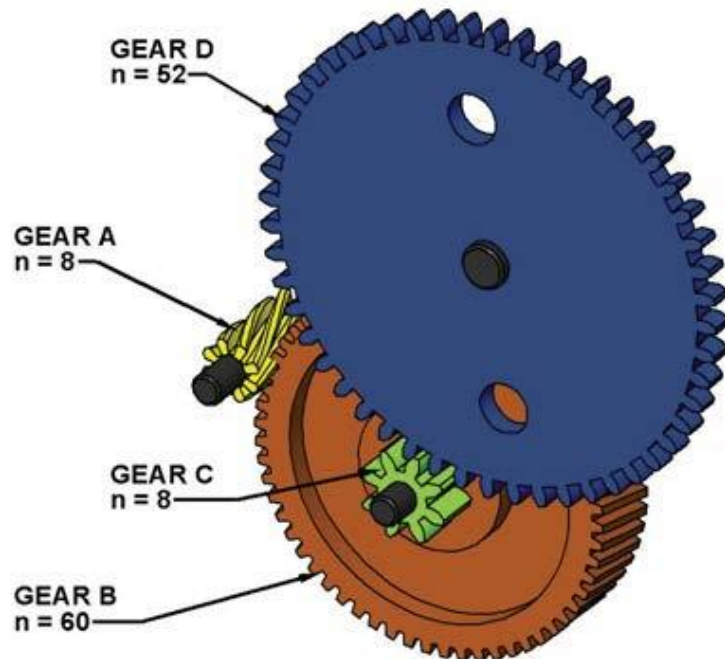
- This gear train transmits motion in the same direction through B idler gear.
- Gear Ratio = $\left(\frac{T_B}{T_A}\right) \left(\frac{T_C}{T_B}\right) = \left(\frac{12}{24}\right) \left(\frac{24}{12}\right) = 1:1$
- Besides direction, torque and speed of A and C will be the same.

Gear Train



- A paired gear rotates in opposite direction each other
- The gear ratio is the same as it for spur gears.
- If Gear 1 is a driver gear
- $$\text{Gear Ratio} = \frac{T_2}{T_1} * \frac{T_4}{T_3}$$

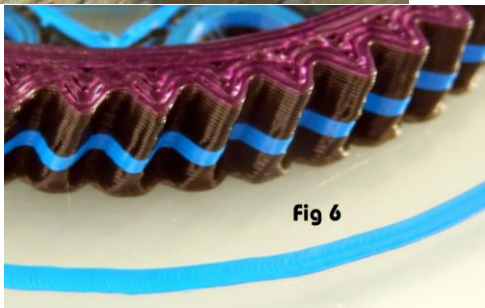
Gear Train



- A is a driver.
- Hence:
- Gear Ratio = $\left(\frac{T_B}{T_A}\right) \left(\frac{T_D}{T_C}\right) = \left(\frac{60}{8}\right) \left(\frac{52}{8}\right) = 48.75:1$
- It means A has to turn 48.75 times to make full revolution of D which will increase D's torque 48.75 fold.

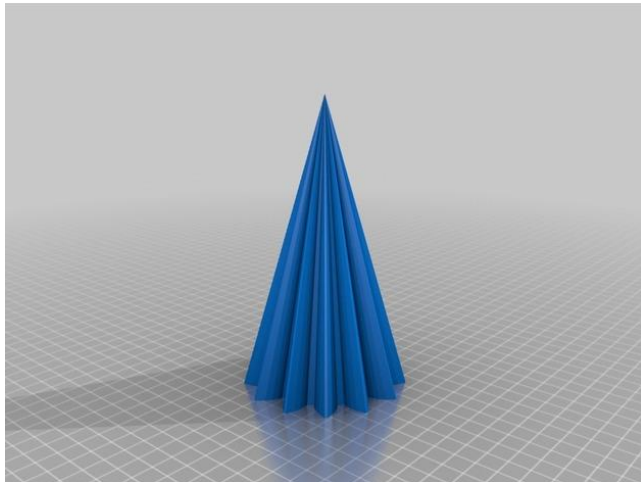
3D Printing Gears

3D Printing Gears Tips



- Holes smaller than 2 mm in diameter shrink
- Small gears with small teeth are challenging, so
 - Module > 0.5 mm
 - Number of Teeth > 12
 - Pitch diameter > 6 mm
- More teeth =>
 - smoother rotary motion and higher contact =>
 - smaller teeth =>
 - poorer printing quality
- Save on material by removing excess parts in big gears

Gear Teeth Tester Model

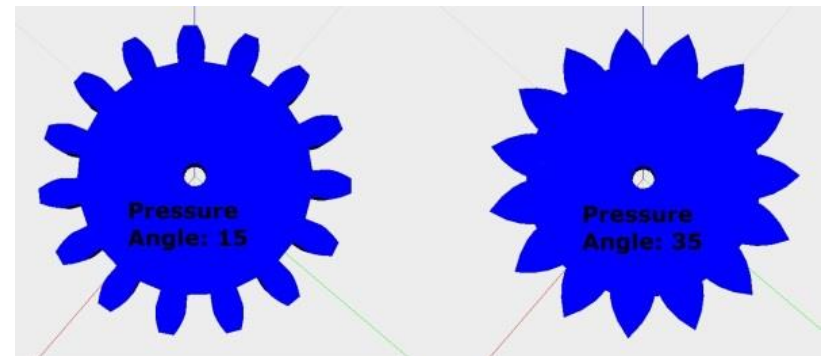


- You can print below tester model to check how well 3D printer can print small gears if you want to use gears with less than 6 mm in pitch diameter.
 - [Tester Model](#)

Gear Teeth

- Mutually Prime Teeth
 - Meshing two gears with >1 common divisor, result in same teeth contacting each other all the time.
 - For example, 15 and 30 same teeth contact.
 - If 15 and 31 is chosen, slightly off gear ratio, but less wear on teeth.

- Pressure angle
 - 20 degrees is the optimal angle for the shape of involute teeth profile.



Lubricate Gear Teeth

- Anything is better than no lubrication
- You can use silicone based oils such as heavy grease, sewing machine oil or any synthetic oil.



Fusion 360

Useful Resources

- Learning Materials
 - [Gears Handbook](#)
- Youtube Courses
 - [All about gears](#)
 - [Gears for 3D printing](#)
- Tools and Online Resources
 - [GF Gear Generator](#)
 - [3D Gears Catalog](#)
 - [Gear Simulator](#)